EXERCISES List 2 October 13, 2026

Simulations and algorithmic applications of Markov chains

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1. For Exercises 8, 9 and 10(b),(c) from List 1, give valid initialization and update functions. For Exercise 10(c), give at least two different choices of initialization and update functions.

2. The total variation distance between probability distributions $\boldsymbol{\mu}, \boldsymbol{\nu}$ on $\mathbb{E}$ is

$$d_{TV}(\boldsymbol{\mu}, \boldsymbol{\nu}) = \frac{1}{2} \sum_{\mathbf{e} \in \mathbb{E}} |\mu(\mathbf{e}) - \nu(\mathbf{e})|.$$

Show that

$$\begin{aligned} d_{TV}(\boldsymbol{\mu}, \boldsymbol{\nu}) &= \sum_{\mathbf{e}: \mu(\mathbf{e}) > \nu(\mathbf{e})} \big(\mu(\mathbf{e}) - \nu(\mathbf{e})\big) \\ &= \sum_{\mathbf{e}: \nu(\mathbf{e}) > \mu(\mathbf{e})} \big(\nu(\mathbf{e}) - \mu(\mathbf{e})\big). \end{aligned}$$

3. Show that

$$d_{TV}(\boldsymbol{\mu}, \boldsymbol{\nu}) = \max_{A \subseteq \mathbb{E}} |\mu(A) - \nu(A)|, \qquad \mu(A) = \sum_{\mathbf{e} \in A} \mu(\mathbf{e}).$$

4. Give a 3×3 transition matrix $\mathbf{P} \neq \mathbf{I}$ having at least two different stationary distributions $\boldsymbol{\pi}$, so that $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$. Can you find a matrix $\mathbf{P} \neq \mathbf{I}$ for which $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$ holds for every probability distribution $\boldsymbol{\pi}$?

5. Let $\mathbf{X} = (X_n)_{n \geq 0}$ be a Markov chain with two stationary distributions $\boldsymbol{\pi}$ and $\boldsymbol{\pi}'$. Show that $p\boldsymbol{\pi} + (1-p)\boldsymbol{\pi}'$ is also stationary for every $p \in (0,1)$.

6. Let (X_k) be a time-homogeneous Markov chain on $\mathbb{E} = \{\mathbf{e}_1, \ldots, \mathbf{e}_M\}$. Show that, for integers $a, a' > 0$ and every $\mathbf{e} \in \mathbb{E}$,

$$\begin{aligned} \mathbb{P}(X_a = \mathbf{e}, X_{a+a'} = \mathbf{e} \mid X_0 = \mathbf{e}) = \mathbb{P}(X_{a'} = \mathbf{e} \mid X_0 = \mathbf{e}) \\ \cdot \mathbb{P}(X_a = \mathbf{e} \mid X_0 = \mathbf{e}). \end{aligned}$$

7. In the Ehrenfest urn model, each of N particles is in container A or B. If $X_k = i$ particles are in A, choose one of the N particles uniformly and move it to the other container. Thus X_{k+1} is $i-1$ or $i+1$. The state space is $\mathbb{E} = \{0, 1, \ldots, N\}$. Give the transition matrix $\mathbf{P}$ and find its stationary distribution.

8. Place each of N particles independently in container A or B, with probability $1/2$ for each. What is the distribution of the number of particles in A?

9. Consider a birth-and-death chain on $\mathbb{E} = \{0,1,\dots,N\}$ with two reflecting barriers. Its nonzero transition probabilities are

$$\mathbf{P}(0,1) = \mathbf{P}(N,N-1) = 1,$$

and, for $1 \le i \le N-1$,

$$\mathbf{P}(i,i-1) = q_i, \qquad \mathbf{P}(i,i) = r_i, \qquad \mathbf{P}(i,i+1) = p_i,$$

where $p_i, q_i > 0$, $r_i \ge 0$ and $p_i + q_i + r_i = 1$. Find its stationary distribution.

Remark. The Ehrenfest model is a birth-and-death chain.

10. Find the stationary distribution of the chain on $\mathbb{E} = \{1,2,3\}$ with

$$\mathbf{P} = \begin{pmatrix} 1-\alpha & \alpha & 0 \\ 0 & 1-\beta & \beta \\ \gamma & 0 & 1-\gamma \end{pmatrix}, \qquad \alpha,\beta,\gamma \in (0,1).$$

11. Show directly from the definition that $\pi(\boldsymbol{\sigma}) = 1/n!$ for every $\boldsymbol{\sigma} \in \mathbb{E} = \mathscr{S}_n$ is the stationary distribution of top-to-random shuffling. Here $\mathscr{S}_n$ is the set of all permutations of $\{1,\dots,n\}$.

12. For top-to-random shuffling of n cards, find the smallest number of steps k_0 that permits a move from any permutation to any other:

$$k_0 = \min\bigl\{k \ge 0 : \ \mathbf{P}^k(\boldsymbol{\sigma}_1, \boldsymbol{\sigma}_2) > 0 \text{ for all } \boldsymbol{\sigma}_1, \boldsymbol{\sigma}_2 \in \mathscr{S}_n\bigr\}.$$

13. Start top-to-random from the identity permutation $(1,\dots,n)$. For $n \ge 3$, show that the distribution at the time k_0 from Exercise 12 is not uniform. For $n = 3$, compute

$$d_{TV}(\boldsymbol{\mu}^{(k_0)}, \boldsymbol{\pi}), \qquad \boldsymbol{\mu}^{(k_0)} = \boldsymbol{\mu}\mathbf{P}^{k_0},$$

where $\boldsymbol{\mu}$ puts mass one at the identity and $\boldsymbol{\pi}$ is uniform on $\mathscr{S}_3$.

14. Let X_0 be the identity permutation of n cards, and let X_1 be the permutation after one top-to-random move. Its distribution is $\boldsymbol{\mu}\mathbf{P}$, where $\boldsymbol{\mu}$ puts mass one at the identity. Compute

$$d_{TV}(\boldsymbol{\mu}\mathbf{P}, \boldsymbol{\pi}),$$

where $\boldsymbol{\pi}$ is uniform on $\mathscr{S}_n$.

15. Let $\mathbf{X} = (X_n)_{n\ge 0}$ be a Markov chain on $\mathbb{E} = \{\mathbf{e}_1,\dots,\mathbf{e}_M\}$ with transition matrix $\mathbf{P}^X$. Assume $\mathbf{P}^X(\mathbf{e},\mathbf{e}) < 1$ for every state $\mathbf{e}$. Construct $\mathbf{Y} = (Y_k)_{k\ge 0}$ by recording only new values of X_n: consecutive

repetitions of the same state are recorded once. For example,

n	0	1	2	3	4	5	6	7	8	9	10
X_n	$\mathbf{e}_1$	$\mathbf{e}_1$	$\mathbf{e}_1$	$\mathbf{e}_2$	$\mathbf{e}_2$	$\mathbf{e}_1$	$\mathbf{e}_3$	$\mathbf{e}_3$	$\mathbf{e}_3$	$\mathbf{e}_2$	$\mathbf{e}_1$

produces $(Y_0,\dots,Y_5) = (\mathbf{e}_1,\mathbf{e}_2,\mathbf{e}_1,\mathbf{e}_3,\mathbf{e}_2,\mathbf{e}_1)$. Show that $\mathbf{Y}$ is a Markov chain and give its transition probabilities in terms of those of $\mathbf{X}$.

16. Let $\mathbf{X} = (X_n)_{n\geq 0}$ and $\mathbf{Y} = (Y_n)_{n\geq 0}$ be independent chains on $\mathbb{E} = \{1,2\}$ with transition matrix

$$\mathbf{P} = \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix}, \qquad \alpha,\beta \in (0,1).$$

Assume $X_0 = 1$, $Y_0 = 2$, and define their first meeting time by

$$T^{\text{indep}} = \min\{k \geq 0 : X_k = Y_k\}.$$

(a) Find the distribution of T^{indep}.

(b) For $\alpha = 3/4$, $\beta = 1/2$ and fixed $\varepsilon \in (0,1)$, find k such that $d_{TV}(\boldsymbol{\mu}\mathbf{P}^k, \boldsymbol{\pi}) \leq \varepsilon$ using

$$d_{TV}(\boldsymbol{\mu}\mathbf{P}^k, \boldsymbol{\pi}) \leq \mathbb{P}(T^{\text{indep}} > k),$$

where $\boldsymbol{\mu} = (1,0)$ and $\boldsymbol{\pi}$ is stationary.

17. Continue Exercise 16, now using common randomness. Again assume $X_0 = 1$, $Y_0 = 2$. Construct a valid update function ϕ and run

$$X_{k+1} = \phi(X_k, U_{k+1}), \qquad Y_{k+1} = \phi(Y_k, U_{k+1}),$$

where $U_1, U_2, \dots$ are i.i.d. uniform on $(0,1)$. The chains are dependent because they use the same update function and random numbers. Let

$$T^{\text{dep}} = \min\{k \geq 0 : X_k = Y_k\}.$$

(a) Find the distribution of T^{dep} for your update function.

(b) For $\alpha = 3/4$, $\beta = 1/2$ and fixed $\varepsilon \in (0,1)$, find k such that $d_{TV}(\boldsymbol{\mu}\mathbf{P}^k, \boldsymbol{\pi}) \leq \varepsilon$ using

$$d_{TV}(\boldsymbol{\mu}\mathbf{P}^k, \boldsymbol{\pi}) \leq \mathbb{P}(T^{\text{dep}} > k).$$

(c) Which meeting time, T^{indep} or T^{dep}, gives the better bound?

(d) Answer the previous question for arbitrary $\alpha,\beta \in (0,1)$.

18. Let $a,b \geq 2$ be integers. Consider a chain on

$$\mathbb{E} = \{(i,j) : i \in \{0,\dots,a-1\},\ j \in \{0,\dots,b-1\}\}.$$

From (i,j) it moves to either $((i+1) \bmod a, j)$ or $(i,(j+1) \bmod b)$, each with probability $1/2$.

(a) Show that the chain is irreducible.

(b) Show that it is aperiodic if and only if $\gcd(a,b) = 1$.